

Hybrid Systems approach to bipedal locomotion with slippage and hybrid control of dispersion.

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Abstract

Foot slip is a major source of instability in bipedal locomotion on low-friction or uncertain terrain. Standard control approaches typically assume no-slip contact and therefore degrade when slip occurs. We propose a control framework that explicitly incorporates slip into the locomotion model through virtual non-holonomic constraints, which regulate the tangential stance-foot velocity while remaining compatible with the virtual holonomic constraints used to generate the walking gait. The resulting closed-loop system is formulated as a hybrid dynamical system with continuous swing dynamics and discrete impact events.

We also discuss a hybrid systems approach to controlling dispersive decay for non-autonomous Hamiltonian systems. While the general theory for dispersion in such non-autonomous systems is largely open, there exists a time-periodically forced one-dimensional Dirac equation with unusually slow dispersive decay rate of $t^{-1/5}$. It is to be expected that such behavior is not generic and requires a very particular forcing term of hybrid type; we provide a more general ansatz and systematic procedure to construct such an equation with a dispersive decay rate no faster than $t^{-1/10}$.

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